

June 23, 2026

Hiring Team, Babel Street (Analytics Group)
NLP/Linguistics Software Engineer
Somerville, MA

Dear Hiring Team,

I'm applying for the NLP/Linguistics Software Engineer role in the Analytics Group. The framing in your posting is what drew me: when a match decision affects whether someone is onboarded or investigated, "the model said so" is not good enough. I have spent my projects building search and ranking systems where every result is measurable, tunable, and defensible, which is exactly what record matching demands.

What I bring to the Analytics Group:

- **Search internals and re-ranking, end-to-end.** VectorLift is a semantic search and ranking engine I built on MS MARCO: a BM25 baseline in Elasticsearch, a fine-tuned dense bi-encoder, and a cross-encoder re-ranker, with Kafka-driven indexing into Elasticsearch and Qdrant served through FastAPI. This is the Elasticsearch data modeling, performance tuning, and retrieval architecture your record matching runs on.
- **Provenance and explainability, by construction.** I evaluated six pipeline configurations with NDCG@10, MRR@10, and MAP, and used paired bootstrap significance tests so each ranking change was defensible rather than anecdotal. My LLM Eval Harness applies the same discipline to text analytics: tracking evidence-quote validity and grounded hallucination rate with McNemar tests and Holm correction. Auditable, tunable matches are the default in how I build.
- **Production Java and Python.** At Isomer AI I shipped 10+ features end-to-end with typed document classification over multi-model LLM pipelines. Independently, my InventoryReservationService is a Java 21 and Spring Boot API with PostgreSQL row-level locking, proven correct under concurrency with Testcontainers integration tests, so I can move comfortably in your core analytics language and your AI/ML integrations alike.

I'm completing my MS in Computer Science at NC State in May 2026, ready to start full-time in summer 2026, and happy to be in Somerville on a hybrid schedule. I'd welcome the chance to help make Babel Street's record matching world-class.

Sincerely,

Aditya Kulkarni